Spatial Gym trajectory (example)

This file inputs `trajectory_fragment.tex` (PNG screenshots from SPaRC-Gym replay and per-step metadata). Regenerate with `python generate_latex_puzzle_trajectory.py` from the repository root.

Spatial Gym trajectory

GPT-OSS 120B (`openai_gpt-oss-120b_gym.jsonl`)

`c2f1726c32030b96`

Difficulty	D2 (score 1.22)
Grid (H×W)	2×2
Outcome	Solved
Logged steps	5 (environment)
Trajectory	5 actions, 6 frames (`step_00` = state before action 1).

Step 1 of 5 *(reasoning below → then action 3)*

Chosen action **Down** (code 3)

Screenshot `step_00.png` = board before this move

Model reasoning

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We need to analyze puzzle.
Grid is 5x5? Represented as rows lines. Coordinates? Current
Position: [0 1] means x=0, y=1? Usually x column, y row.
Starting at coordinate (0,1). Let's map grid with coordinates
(x from 0 left to 4, y from 0 top to 4). Grid rows given:
Row0: ['+', '+', '+', '+', '+']
Row1: ['S', '*-Y', '+', '*-Y', '+']
Row2: ['+', '+', '+', '+', '+']
Row3: ['+', '*-K', '+', '*-K', '+']
Row4: ['+', '+', 'E', '+', '+']
Symbols: S at (0,1).
E at (2,4). stars at (1,1) color Y, (3,1) Y, (1,3) K, (3,3) K.
Current Position is [0 1] which matches S. Legal actions:
up or down. Means cannot go left or right because maybe out of
bounds? From (0,1), up goes to (0,0) which is '+'. Down goes
to (0,2) which is '+'.
Goal: Find a path to E covering all dot cells (none) and obey triangle (none) etc. Rule: cannot
pass through rule cells (stars) since they are gaps? Stars are rule cells at odd coordinates
(both odd). Indeed coordinates (1,1), (3,1), (1,3), (3,3) are rule cells; path cannot go through
them.
We need to find a path from S to E without crossing those star cells, cannot revisit, and must end
at E (2,4). Also there may be region rules: squares none, triangles none, poly shapes none.
Only stars: each star must be paired with exactly one other element of same color within its
region. Stars of color Y at (1,1) and (3,1). Stars of color K at (1,3) and (3,3). They are each
color with two stars; they need to be paired with exactly one other element of same color inside
same region. The other element could be maybe another star? Actually "each star symbol MUST be
paired with exactly ONE other element of the same color." Could be another star? Probably star
pairs themselves count? Not sure. Usually in puzzles like 'Star Pairing', each star must be
paired with a dot of same color. But here maybe another element (could be dot, square, etc.). No
other same color elements exist besides the stars themselves. So each star must be paired with
the other star of same color, but they are separate stars. "Each star symbol must be paired with
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exactly ONE other element of the same color. Other colors within region are irrelevant." So if there are two Y stars in same region, they could pair with each other; each star paired exactly one other element (the other star). That satisfies rule if they are in same region. But if they are separated by path into different regions, they'd each lack a partner. So we need to ensure that stars of same color are in same region. So the path must not separate Y stars from each other, same for K stars.
Thus we need to draw path from S to E that encloses a region containing both Y stars together, and another region containing both K stars together, or maybe they can be in same region as each other? But if Y star paired with K star? No, must be same color, so not.
Simplify: we can make a path that goes around all four star cells maybe creating a region that contains them all together (both colors). That would put Y stars with K stars together, but each Y star still has partner? It could be the other Y star in same region (still there). So okay.
But path cannot go through them, can go on surrounding '+' cells.
We need to find any path from S (0,1) to E (2,4) that doesn't cross star cells and doesn't revisit. Let's attempt to design moves.
Grid visualize with coordinates (x,y):
(0,0) + ; (1,0)+ ; (2,0)+ ; (3,0)+ ; (4,0)+
(0,1) S ; (1,1)*-Y ; (2,1)+ ; (3,1)*-Y ; (4,1)+
(0,2)+ ; (1,2)+ ; (2,2)+ ; (3,2)+ ; (4,2)+
(0,3)+ ; (1,3)*-K ; (2,3)+ ; (3,3)*-K ; (4,3)+
(0,4)+ ; (1,4)+ ; (2,4)E ; (3,4)+ ; (4,4)+
We can move through any '+' cells.
We start at (0,1). Allowed moves up to (0,0) or down to (0,2). Need to eventually go to (2,4).
Let's think simple path: go down, right, down, right, down, right? But must avoid stars at (1,3) and (3,3). Also (1,1) and (3,1) are stars.
Potential path: from S down to (0,2), then down to (0,3), then down to (0,4), then right to (1,4), right to (2,4) reaching E. This path would be: S (0,1) -> (0,2) -> (0,3) -> (0,4) -> (1,4) -> (2,4) E. This does not cross any star cells. Does it satisfy star pairing region rules? The path will create only one region (outside) and maybe interior region? The path is a simple line that does not enclose any area because it's not closed loop; path divides plane but region concept maybe only closed loops create enclosed areas. But stars are not in the path, they are on the other side of path. For star pairing, they need to be in same region; path does not partition them separate as they are on same side (since path is on left side of them). All stars are to the right of path (since path stays at x<=0 then x=1 at (1,4) but still left of stars). Since they stay together, they are in same region (the region not occupied by path). So they are paired within same region. So rule satisfied.
Thus this simple path seems valid. No dot constraints.
Thus first move: we must choose from legal actions: up or down. Down leads to (0,2) which continues as described. Up would go to (0,0) maybe also possible but then need to navigate. Both may lead to solution.
Pick down (3). So final answer digit 3.
We must output "Final: 3".

Step 2 of 5 *(reasoning below → then action 3)*

Chosen action **Down** (code 3)

Screenshot step_01.png = board before this move

Model reasoning

We need to decide next move. Let's interpret grid. Coordinates? Current position [0 2] means x=0, y=2? Grid indexing probably row,column?
Given grid displayed as rows. Row0: ['+', '+', '+', '+', '+']
Row1: ['V', '*-Y', '+', '*-Y', '+']
Row2: ['L', '+', '+', '+', '+']
Row3: ['+', '*-K', '+', '*-K', '+']
Row4: ['+', '+', 'E', '+', '+']
Current position L at row2 col0. Legal actions: [0=RIGHT, 3=DOWN]. So can go right to col1 row2 or down to row3 col0.
Goal is to reach End E at row4 col2. Must also pass through any dots '.' (none visible). Must avoid gaps G (none). Must go around rule cells squares, stars etc. We have stars at positions (row1 col1) and (row1 col3) color Y. Also stars at (row3 col1) and (row3 col3) color K. Triangles? none.

Squares? none. Need to satisfy region constraints: each region squares same color; not relevant. Stars: each star must be paired with exactly one other element of same color within same region. That element can be? Probably another star of same color? Or maybe something else? Rule says each star must be paired with exactly ONE other element of same color. Other colors irrelevant. Likely need to have stars of same color paired together within same region. So Y stars must be paired together, K stars together. Must be in same region (enclosed area separated by path). So path must separate Y and K stars maybe each region contains both Y stars? Actually they are currently separate cells: Y stars at (1,1) and (1,3). K stars at (3,1) and (3,3). Path can separate them into regions such that each region contains exactly two stars of same color perhaps? The rule: each star must be paired with exactly one other element of same color. So each Y star must pair with another Y (the other Y star). So both Y stars should be in same region. Similarly K stars in same region. Therefore path must not separate Y stars from each other; same for K.
Currently path started at some start node at maybe top-left? We have visited cell V at row1 col0, indicating path from start has moved down? Let's track: Starting location? Not given, but currently visited V at row1 col0. L at row2 col0. So path goes from start somewhere above row0 maybe? Probably start at (0,0) top-left? Actually grid top-left row0 col0 is '+'. Possibly start S is at that location but not shown because visited? Wait V at row1 col0 indicates that cell was visited earlier. So start maybe at row0 col0? Then moved down to row1 col0 (now V), then down to row2 col0 (now L). So path downwards column0.
Goal is to reach E at (4,2). Need to navigate around stars. Since stars are at (1,1) (1,3) (3,1) (3,3). Path cannot go through them because they are rule cells (odd coordinates both). Indeed they are '*' entries. Path can go around them.
Current path column0. Options: Right to column1 row2, which would move into a cell currently '+' at row2 col1. That cell is adjacent to Y star at (1,1) (above) and K star at (3,1) (below). Passing through there will affect region boundaries.
Alternatively down to row3 col0 which is '+' (currently). Let's consider overall plan: need to end at (4,2). Likely path will go down to bottom then right to column2. Must also ensure stars are paired properly: both Y stars must be in same region. If we draw a path that separates top row Y stars from each other? They are separated by column2 (col2 is '+'), but they are at col1 and col3 with a plus cell between. Path can go between them to separate them into different regions. To keep them together, we need to ensure path does not cut between them. Similarly for K stars.
Thus, we should avoid creating a wall that cuts between the two Y stars horizontally across row1 between col1 and col3. That would be at row1 col2, which is currently '+'. If path passes through row1 col2, it would separate Y stars into different regions (above? Actually path goes through that cell; the region on one side might separate). Hard to reason.
Given start column0, maybe we go down to row3 col0 then right across bottom row, then up etc. Let's sketch potential path: currently at (2,0). Down to (3,0), then down to (4,0), then right to (4,1), right to (4,2) which is E. That would be straightforward, not intersecting any stars. Does any rule forbid passing adjacent to stars? No.
Check legal actions: after moving down from (2,0) to (3,0), would there be legal moves? Need to see if path passes through any '.' none.
Thus probably best move is DOWN (digit 3). Let's ensure not revisiting: (3,0) is not visited yet. No constraints broken.
Thus final decision digit 3.

Step 3 of 5 *(reasoning below → then action 3)*

Chosen action **Down** (code 3)

Screenshot `step_02.png` = board before this move

Model reasoning

We have a path-finding puzzle. Need to decide next move. Step 3, current position [0 3]. Coordinates likely x y? Grid representation: rows? They show grid as 5x5. Indices maybe [x y] means column x, row y? Starting at (0,3) which is first column, fourth row (0-index). Grid shows row order: first row ['+', '+', '+', '+', '+']; second row ['V', '*-Y', '+', '*-Y', '+']; third row ['V', '+', '+', '+', '+']; fourth row ['L', '*-K', '+', '*-K', '+']; fifth row ['+', '+', 'E', '+', '+']. At current position L at row4 column0 (0-index). Adjacent moves: Right would go to '*-K' (a star) which is a rule cell (odd coordinates) maybe not allowed for path (path cannot

go onto rule cells). According to core concept: Path travels only on positions marked '+' or '.' . In grid, '*' cells are rule cells - path can't go on them. So right is not legal. Up would go to row3 column0 which is 'V' (visited). Cannot revisit? Path cannot revisit cells. So up not legal. Left would go out of bounds. Down goes to row5 column0 which is '+' (bottom row). Legal actions list only 3=DOWN. So must go down. So final answer digit 3.

Step 4 of 5 *(reasoning below → then action 0)*

Chosen action **Right** (code 0)

Screenshot step_03.png = board before this move

Model reasoning

We need to
decide the move. Let's understand grid layout. Coordinates: current position [0 4] meaning x=0, y=4? Probably (col, row) with origin top-left (0,0). Grid shown rows from top (row0) to bottom (row4). Row0: ['+','+','+','+','+'] columns 0-4.
Row1: ['V','*-Y','+','*-Y','+']
Row2: ['V','+','+','+','+']
Row3: ['V','*-K','+','*-K','+']
Row4: ['L','+','E','+','+']
Current position L at row4 col0. End E at row4 col2.
There's a plus at row4 col1 between L and E. Also gap? no.
Legal actions: [0=RIGHT]
only. So must move right to column1 row4. That's allowed.
Check other constraints: path cannot revisit,
and must pass through all dots (none). Should be okay.
Thus final action 0.
But ensure cannot go through gap? The cell at row4 col1 is '+', not gap.
Thus answer.

Step 5 of 5 *(reasoning below → then action 0)*

Chosen action **Right** (code 0)

Screenshot step_04.png = board before this move

Model reasoning

We need to understand the puzzle. Grid
is 5x5 rows? Let's map coordinates. Provided grid rows list:
Row0: ['+', '+', '+', '+', '+']
Row1: ['V', '*-Y', '+', '*-Y', '+']
Row2: ['V', '+', '+', '+', '+']
Row3: ['V', '*-K', '+', '*-K', '+']
Row4: ['V', 'L', 'E', '+', '+']
Current position [1 4] presumably (x, y). Need to confirm coordinate system: likely (x column, y row) zero-index?
Starting with positions like row indices. The current position is at row4, column1? Actually [1 4] meaning x=1, y=4. Row index 4 is bottom row (row4). Column index 1 within that row is element 'L' (makes sense). So L is at row4 col1.
Grid includes 'E' at row4 col2 (x=2,y=4). So end
node is adjacent to current position horizontally to the right.
Legal actions list: [0=RIGHT] only move possible (maybe no up etc). So we must move right to reach E. The path will go from L at (1,4) to E at (2,4). That would end puzzle.
We must output Final: 0 (RIGHT).
We should ensure that moving right is allowed (not a gap or visited). Row4 col2 currently 'E' which is end; moving into E is allowed, I think. So final step.
Thus answer: Final: 0

Final frame *(after last move: **Right**, code 0)*

`step_05.png` *terminal state*